



CISC/CMPE 320
Fundamentals of Software Development
Fall 2026 Syllabus

In addition to syllabus in this document, students must read the CISC “Common Syllabus Information” page at <http://www.cs.queensu.ca/students/undergraduate/syllabus/>. That document contains important administrative information that applies to all CISC courses, and in turn refers you to a few important pages on the Queen’s website which are required reading. Please read all of those documents carefully. This information is essential for you as it clarifies your rights and responsibilities. It is a must that all students be familiar with all of these provided information.

Contact Instruction:

To communicate with the instructor. Please include the following suffix in your email “320_ABCD_Topic” where the “320” stands for the course number. The “ABCD” represents last four digits of your student ID. Please insert “_” (Underscore) between the three different parts of your subject.

For example:

Your student ID # is 20143670 and you sending an email regarding “Assignment 2” then the subject of your email should be:

320_3670_Assignment 2

Instructor:

Dr. Samir Mohammad

Office: Robert Sutherland Hall, Rm 532

Office hours: by appointment

Tuesdays 2 – 3

Wednesday 12 – 1

Email: sm373@queensu.ca

Teaching Assistance:

Name	Email
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Office Hours: During the Tutorial, feel free to ask questions.

Lecture Time:

Mondays	4:30 – 5:30
Wednesdays	3:30 – 4:30
Fridays	2:30 – 3:30

Lecture Location:

Dupuis Auditorium

Tutorial Venue : Kingston, Rm 209

Tutorial Time	: Mondays	9:30 – 11:30
	Thursdays	9:30 – 11:30
	Thursdays	12:30 – 2:30

Pre- or Co-requisite:

Registration in a School of Computing Plan and a minimum grade of a C- (obtained in any term) or a 'Pass' (obtained in Winter 2020) in CISC 235.

Course Description:

According to the [TIOBE Programming Community Index](#) the programming languages C and C++ were the top two languages in use in 1999. Since then, they have fallen to positions 1 and 4 (Java is #2, Python is #3). If the usage were combined, C+C++ (equals C+++?), then they would again take the top spot. C++, which evolved from, and contains C, is still the language of choice when performance and a small memory footprint are desired. While C++ is primarily an Object-Oriented language it can still contain procedural code and has been criticized for being overly complicated - making the language more difficult to learn and use safely. C++ libraries and compilers abound, and several different extensions of the C++ languages are in use, such as C++/CLI in Microsoft's .NET framework.

In this course you will learn how to program in ISO C++, learn to build GUIs using Qt Creator, Codeblocks or C++/CLI and will work on a C++ project as part of a small team. *This will all look great on your resume!*

Course topics can be split into three major areas: ISO Standard C++, C++/CLI and Software Engineering. It is difficult to predict how many course topics will be covered, in what order, and to what depth. It might be best to consider the following as a "wish list"!

ISO Standard C++

Given your background in Python and/or Java, some fundamental topics like conditionals and loops will only need a short review, where the emphasis will be on any differences in the C++ implementations. Also, C++ is not a "safe" language like Java - opportunities for creating problems abound! Such C++ "Gotchas" will be emphasized whenever possible.

- The official implemented C++ standard: C++14
- Fundamental Types and Declaration
- Expressions and Type Casting
- Console Input/Output
- Libraries and Namespaces
- Pre-processor Directives
- Conditionals and Loops
- Functions
- Scope Rules
- Pointers and References
- Arrays and Vectors
- Structures
- Classes
- Constructors
- Destructors
- Operator Overloading
- Strings
- Streams and File I/O
- Inheritance
- Virtual Functions and Polymorphism
- Templates
- Exceptions
- The Standard Template Library or "STL"

C++/CLI

CLI stands for "Common Language Infrastructure" which is a Microsoft language that extends ISO C++. It is implemented in Visual Studio .NET. In .NET you can combine Visual Basic, C++ and C# code to form one program through the use of a platform independent language called MSIL or "Microsoft Intermediate Language". You can write ISO standard C++ in .NET, but then you will not be taking advantage of what this framework offers. Visual Studio and .NET provide a consistent, well documented language interface, a visual editor that sets the standard for all visual editors, and allows you to write managed code that is safer than normal ISO C++. Of course, Microsoft is not terribly interested in allowing you to create multi-platform code...

- Differences in Basic Syntax
- gcnew Versus new
- Handles as Compared to Pointers
- Managed Code Compared to Unsafe Code
- Building and Using *.dll Libraries
- Generics
- Properties, Delegates and Events
- .NET Framework Libraries
- Integrated XML Documentation
- Windows Form Applications

Software Engineering

These topics are designed to aid team software development and to break up the monotony of just talking about C++ programming. Hopefully we can discuss topics early enough so that you will have a chance to try them out in your own project. Much of software engineering is applied common sense, but you need to learn the jargon and some standard techniques so that you can better communicate with the rest of your team and the programming community at large.

- Dealing with Complexity
- UML Diagrams
- Version Control
- Code Repositories
- Testing and Test-Driven Development
- Bug Tracking
- "The Mythical Man Month"
- Team Communications and Documentation
- Agile Programming Techniques

Textbook:

- Recommended textbook: C++ How to Program, Paul Deitel and Harvey Deitel. 10th Edition.
- More Valuable References:
 - <https://www.learncpp.com/>
 - <http://www.cppreference.com/>

Course Objectives:

By the end of this course, students will:

1. Design and construct medium-sized console and GUI programs in both procedural and object-oriented paradigms.
2. Design and construct 500 to 1000 line console and GUI C++ programs using both Object Oriented and procedural C++ code.
3. Contribute to the design and construction of a larger software project as a member of an agile programming team.
4. Use software engineering techniques and tools to aid in the development of larger scale software projects.

Weekly schedule:

Week	Of	Subject	Assignments Due 23:59 on Wednesdays	Projects Due Fridays
1	Sep 8 (Tu)	Introduction, Setup, History, Basics		
2	Sep 14	Data Types, Pointers, Arrays		
3	Sep 21	Agile, Passing, Operators, Loops, String	Assign 1 due (Sep 23, 11:59 pm)	
4	Sep 28	I/O, Classkey, Exceptions, RAD SCRUM		Project Outlines (Oct 2, 11:59 pm)
*5	Oct 5	Functions, Overloading 1	Assign 2 due (Oct 7, 11:59 pm)	
Oct 12 - 18 Reading Week				
6	Oct 19	Overloading 2, Overloading 3, Repositories, SDD		
7	Oct 26	SW Analysis 1, SW Analysis 2, SW Design.		RAD Submission (Oct 30, 11:59 pm)
8	Nov 2	Namespaces, Memory, Memory Errors, Help Tools		
9	Nov 9	Inheritance 1	Assign 3 due (Nov 11, 11:59 pm)	SDD Submission (Nov 13, 11:59 pm)
10	Nov 16	Inheritance 2, Inheritance 3, Inheritance 4, Testing.		
11	Nov 23	Testing 1, Testing 2, Templates, STL	Assign 4 due (Nov 25, 11:59 pm)	
12	Nov 30	Smart Pointers		Project Presentation
13	Dec 7	Just Monday of this Week		

Marking Scheme

- 4 Assignments, 7.5% each, Total 30 %
- Project 35 %
- Final Exam, 35 %
- Total: 100%
- Your overall percentage (from the scheme above) will be rounded to the nearest whole number and converted to a letter grade according to the rules in Section 10 of the Arts & Science Calendar, as follows:

90-100% = A+ Exceptional	63-66% = C Minimally Acceptable (Honors)
85-89% = A Outstanding	60-62% = C- Minimally Acceptable (General)
80-84% = A- Excellent	57-59% = D+ Unsatisfactory Pass
77-79% = B+ Very Good	53-56% = D Unsatisfactory Pass
73-76% = B Good	50-52% = D- Unsatisfactory Pass
70-72% = B- Reasonably Good	0 - 49% = F Failure – No Course Credit
67-69% = C+ Acceptable	

Assignments Late Submission Policy

All assignment will be given one week to complete, which constitute the original given time. Then another 2nd week would be given if for some reason you couldn't complete the assignment within one week. For example, if you get a flu in the 1st given week, then you may rest for this week and complete your assignment during the 2nd week. This privilege is to be used responsibly only for emergencies. If your submission within these 2 weeks, then no further action is required. And please don't take it as all assignment's deadlines are in 2 weeks. That is not correct. The deadline for all assignments is one week. And the 2nd week is given for emergencies.

If you pass the original deadline within 7 days, and then also pass the emergency grace period of extra week (the 2nd week) then your submission is subject to be investigated and action taken will depend on one of the following 3 cases:

- 1- Students with **letters of accommodation** should, if possible, confirm the implementation of their accommodations prior to the listed due date. Please see the "Accommodations for Disabilities" section of this syllabus for more information.
- 2- Students experiencing short-term extenuating circumstances that are beyond their control and may affect their academic work should submit a request to their faculty office for academic consideration. Please see the "**Academic Considerations** for Students in Extenuating Circumstances" section of this syllabus for more details.
- 3- **Urgent Matters** to be directed to the instructor directly will be granted up to maximum 7 days, such that the applicant should provide an academic consideration for students in extenuating circumstance as soon as possible.

Late Policy:

Late assignments beyond 2 weeks (without long-term Considerations or Accommodations):

Anything that arrives after this will not be accepted without permission of the instructor.

Please see the Academic Considerations for Students with Extenuating Circumstances and Accommodations for Disabilities sections of the syllabus for more information.

Location and Timing of Final Examinations

The exam dates for each Term are listed on the Faculty of Arts and Science webpage under "[Important Dates](#)." Student exam schedules for the Fall Term are posted via SOLUS immediately prior to the Thanksgiving holiday; for the Winter Term they are posted on the Friday before Reading Week, and for the Summer Term they are individually noted on the Arts and Science Online syllabi. **Students should delay finalizing any travel plans until after the examination schedule has been posted. Exams will not be moved or deferred to accommodate employment,**

travel/holiday plans or flight reservations. Also, as indicated in Academic Regulation 8.3, students must write all final examinations in all on-campus courses on the Kingston campus.

Assignments:

Students will be given 4 marked assignments. These assignments are essential for doing well in the course as they help to prepare you for the tests among other resources.

OnQ Online help:

Course schedule will be available on OnQ by the start of the term. Students will have adequate warning of all assignment due dates. In addition, all important information will be announced in regular basis. Therefore, students are advised to check the course's OnQ website regularly.

Students with Temporary or Permanent Needs:

If you are unable to complete an assignment or write a quiz due to illness or other exceptional circumstances, please refer to the Academic Considerations and Accommodations on the School of Computing web site (<http://www.cs.queensu.ca/students/undergraduate/syllabus/#extenuating>).

Acknowledgement of Territory:

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory.